MATHEMATICS

The Honest Coder

Maths trick #5

Website → https://thehonestcoder-edu.github.io/

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Trick: Square Root of numbers upto 4 digits

Step by Step Explanation

(with examples)

1. Divide the number into two parts - last 2 digits and remaining 1-2 digits at the beginning.

$$65 | 61$$

2. Find the highest perfect square lesser than Group 1 (first 1-2 digits). This is the first digit.

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3. Find the digits whose squares end in the units digit of Group 2 (last 2 digits). In most cases, 2 of the 10 digits would have the same number. Either of these 2 digits will be the last digit of the square root.
4. Multiply the first digit of the square root (we obtained this in step 2!) with the number that comes immediately after the digit.
5. If group 1 is greater than the product, than the bigger digit of the 2 options is the right answer, else the smaller digit is the last digit.

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NOTE - If the procedure felt too complex, dont worry, the examples will help make sense.

Example 1→

$\sqrt[2]{8464}$

Step 1 - Make two groups -

- Group 1: first 2 digits (84)
- Group 2: last 2 digits (64)

Step 2 - The biggest perfect square smaller than 84 is 81, which is the square of 9.

Step 3 - The only digits whose square end with 4(the last digit of group 2) are 2 and 8. Therefore the second digit is either 2 or 8.

Step 4 - 9 * 10 (10 comes just after 9)

→ 90

Step 5 - Since 84 < 90, therefore, 2 is the right last digit.

Final answer → 92

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Example 2→

$\sqrt[2]{3249}$

Step 1 - Make two groups -

- Group 1: first 2 digits (32)
- Group 2: last 2 digits (49)

Step 2 - The biggest perfect square smaller than 32 is 25, which is the square of 5.

Step 3 - The only digits whose square end with 9(the last digit of group 2) are 3 and 7. Therefore the second digit is either 3 or 7.

Step 4 - 5 * 6 (6 comes just after 5)

→ 30

Step 5 - Since 49 < 30, therefore, 7 is the right last digit.

Final answer → 57

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Example 3→

$\sqrt[2]{4225}$

Step 1 - Make two groups -

- Group 1: first 2 digits (42)
- Group 2: last 2 digits (24)

Step 2 - The biggest perfect square smaller than 42 is 36, which is the square of 6.

Step 3 - The only digit whose square ends with 5(the last digit of group 2) is 5. Therefore the second digit is 5.

Final answer → 65

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Practice sheet

1. $\sqrt[2]{1225}$
2. $\sqrt[2]{1764}$
3. $\sqrt[2]{5476}$
4. $\sqrt[2]{4624}$
5. $\sqrt[2]{8649}$

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